

Aspect-Based Sentiment Analysis

Predicting Aspects

NLP AI Hackathon

This project aims to **predict aspects** of reviews and analyze their sentiment, achieving an impressive **94% accuracy**. Aspect-based sentiment analysis is crucial for understanding customer feedback, allowing businesses to respond effectively and enhance their services based on insights gained from reviews.

[Presenter's
First Name]

Data Pre-processing Steps

Clean Data

Raw review data was cleaned to remove irrelevant information and ensure clarity for further analysis.

Split Data

The dataset was divided into training, validation, and test sets to effectively evaluate model performance.

Drop Duplicates

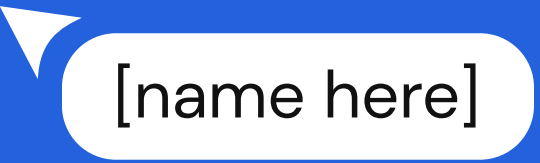
Duplicate entries were removed to maintain data quality and prevent skewed results during model training.

Normalize Text

Text normalization was performed to ensure consistent input across reviews, improving model comprehension and accuracy.

Understanding Language Models

Employed XLM-RoBERTa (XLMN) masked language model to enhance **contextual comprehension** and improve domain-specific language understanding.



[name here]

ENSEMBLE MODEL COMPONENTS

Overview of the Trained Models Used for Analysis

COMBINING STRENGTHS

The Ensemble Approach

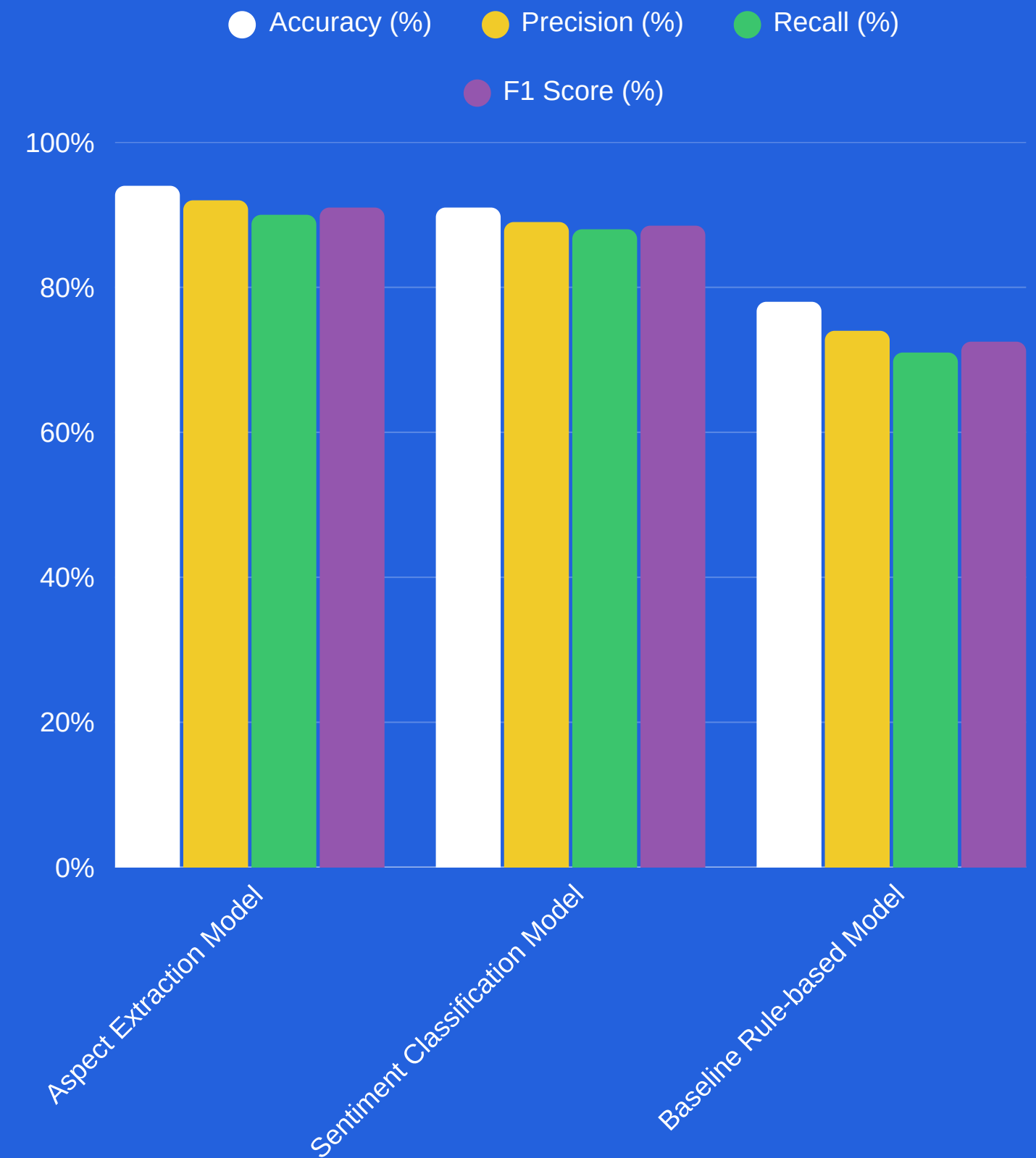
We **combined outputs** of MarBERTv2, CAMeLBERT, and XLM-RoBERTa using an ensemble method to achieve improved accuracy and better performance than single models.

Project Outcomes

94% Accuracy

High Performance

Consistent on Test Dataset



Source: AI-generated data. Replace or verify before use.

Key Takeaways from the Project

Data Quality

Ensuring high data quality and normalization significantly impacts model performance and reliability in predictions.

Unlabeled Data

Utilizing unlabeled data with masked language models enhances the model's understanding of various contexts and nuances.

Ensemble Techniques

Applying ensemble techniques improves the robustness of predictions and increases overall accuracy across models.

Multi-lingual Models

Implementing multi-lingual models enhances adaptability and effectiveness across diverse languages and cultural contexts.

Thank you!

Feel free to reach out for questions or further discussions about the project.